

SYLLABUS FOR PRECLINICAL STUDIES

MODULE I

BODY STRUCTURE, ORGANIZATION AND FUNCTIONS (BSOF 1011)

MODIFIED December 2022

- I. Module title:** Body Structure, Organization and Function
- II. Module code:** BSOF 1011
- III. Credit value (ECTS):** 12
- IV. Duration:** Year I, Semester I
- V. Prerequisites:** Premedical courses
- VI. Module Description**

This module is designed for first year undergraduate medical students and is intended to describe body structure and function, as well as molecular mechanisms that operate in the living cell. It also deals with developmental periods, with an emphasis on the beginning of human development, formation of embryonic discs and organogenesis, the fetal period, human birth defects, formation of all body systems and molecular regulation of development.

VII. Module Contents and Learning Outcomes

Module objectives:

At the end of the module the preclinical students will be able to:

- be acquainted with anatomy, physiology and biochemistry related medical terminologies
- recognize the body's structure, organization and function, including molecular mechanisms that operate in the living cell
- dissect cadavers, use and operate light microscopes, learn basic histological techniques and perform biochemical tests in the living system
- recognize that physiology is a science involving adjustment mechanisms to prevailing environmental conditions and understand the concepts of homeostasis, allostasis and heterostasis

Supportive objectives of the Module

At the end of the module the preclinical students will be able to:

- differentiate normal and abnormal anatomical variations
- explain the distribution of fasciae in the body

- interpret the potential spaces in the body cavities and the arrangement of their membranes
- describe the structure and functions of skin, superficial fascia and subcutaneous tissue
- compare the processes of bone formation and resorption
- explain the components of the skeletal muscles of the back and upper limb
- explain the organization of the nervous system, the spinal cord and nerve roots
- interpret the distribution of the main blood and lymphatic vessels in the body
- adhere to regulations of the dissection room
- follow the guidelines for dissection of the back and upper limb regions
- trace the proximal and distal attachments, innervations and vascularization of each muscle in the back and upper limb too
- trace the vertebrae, curvatures, ligaments and joints of the backbone
- describe the sites of safe lumbar puncture
- describe common bone fractures and their management
- identify the superficial, intermediate and deep back muscles, along with their blood and nerve supply
- identify the anterior and posterior compartment muscles of the upper limb, along with their blood and nerve supply
- acquire basic microscopy skills
- observe how tissue sections are prepared for light microscopic examination
- explain the basic concepts of cytology and histology
- identify the organelles, inclusions and cytoskeletal system of the cell
- identify the molecular biology of the cell and the relationships between the organelles, cell inclusions and the cytoskeletal system with emphasis on cell cycle and interactions of the organelles.
- describe the organization of cells and extracellular matrix in the tissues, organs and organ systems
- explain the basic concepts of embryology
- demonstrate knowledge of the sequence of development from fertilization, formation of tissue organs and organ systems
- identify developmental malformations of organs and organ systems.
- demonstrate understanding of the interaction of cells for induction, determination and differentiation in the formation of organs and organ systems, as well as molecular regulation for such interactions
- describe genetic and environmental factors that govern the development of the human body
- describe the basic principles of physiological adjustment.
- explain the molecular basis of membrane structure, cellular organelles and their functions
- describe the distribution of fluid and electrolyte across cell membranes
- describe transport mechanisms across cell membranes
- describe the autonomic nervous system
- understand the bioelectricity of cells
- explain autonomic control of visceral function

- demonstrate the chemical basis of life
- describe the role of water in the cell and acid-base properties
- describe the biological chemistry of the various biomolecules and assess their role in various cellular processes
- describe how biomolecules are organized to perform their cellular functions
- provide basic information on the structure and function of genetic material

VIII. Learning and Teaching Strategies

8.1. The strategies include:

- Small group learning
- Practical learning
- Demonstrations

8.2. Methods of Teaching and Learning:

- Standard Lecture
- Seminars
- Tutorials
- Demonstration of Techniques
- Practical learning

NB: In this module the instructional strategies for physiology will include only lectures and tutorials/seminars.

8.3. Human resources:

- **Staff:** Instructors and graduate assistants drawn from the department of anatomy, physiology and biochemistry will be involved in lecture delivery, laboratory demonstration, facilitation and assessment of seminars, tutorials, quizzes and tests.
- **Spaces/location:** AAU, CHS, lecture halls (**SefereSelam**), labs, dissection rooms (**Black Lion**)
- **Instructional media:** interactive whiteboard, printed materials (handouts, books, and worksheets), blackboards, display board, flip charts, slides, overheads, real objects or models, images, projector, LCD projectors, DVDs and CD-ROMs will be used during lectures, seminars and tutorials. The laboratory session requires, pH meters and solutions of various pH strengths for biochemistry; forceps, scissors, blades, wood block, dissection table etc for anatomy; microscope, images for histology; preserved embryonic specimens for embryology.

- **Time:**Lecture : Monday to Friday from 9:00-11:50a.m.
Laboratory: Monday to Friday 1:30 – 04-30 p.m.

IX. Role of Instructors:

The role of instructors includes giving standard lectures, seminars and assignment; involving learners fully in assessments, help students to understand what is expected, addressing students questions, facilitating discussion, and developing learners roles in moderation activities. It is also to instructors' role to plan, design and carry out the lesson plan, to collaborate with physiology and biochemistry staff to accomplish the module successfully, to demonstrate prosected cadaver anatomy, to evaluate students and to correct quizzes, tests and exams.

X. Role of learners:

It is the learners' role to engage actively in learning, attend lectures, participate actively, share ideas with each other, ask questions on unclear issues, do seminars and assessments, choose topics for tutorials, refer to different reading materials, review their own learning through self assessment and to demonstrate their knowledge, skill, attitudes and capabilities through a range of evidence including self-assessment, quiz, tests, activities, seminars and examination.

XI. Assessment methods

Assessment criteria of anatomy, embryology, histology, physiology and biochemistry

- Seminar/assignment - 10%
- Two tests – 50%
- Final examination- 40%

NB: The above assessment methods are open for changes.

Final
Schedule

Module one part one, course calendar (Week 1-5)

February 7 – Mar 11

Module	Coordinator	Department
Module name: Body structure, Organization and Functions -1 Module number: BSOF 1011 ECTS: 6	Shewit Gebre (MSc)	Anatomy
	Abebye Aragaw (PhD)	Physiology
	SisayAddisu (PhD)	Biochemistry
Placement: SefereSelam Campus (for lecture)		

Schedule Module BSOF 1011

MODULLE ONE Part I: (Body structure organization and function =BSOF 1011)					
THEME ONE: Anatomy					
TIME TABLE FOR WEEK ONE (Dec5 – 9)2022					
TIME	MONDAY (Dec 5)	TUESDAY (Dec6)	WEDNESDAY (Dec 7)	THURSDAY (Dec8)	FRIDAY (Dec9)
9:00:-10: 30 a.m. INSTRUCTOR	Definition, history and approaches of studying anatomy Anatomicomedical terminologies --- Directional terms --Body planes -- Anatomical variations Shewit Gebre	Over view of Skeletal system; Axial skeleton bones and joints Appendicular skeleton bones and joints Over-view of Articular system; Classification of joints Shewit Gebre	Organization of the Nervous System: Cells CNS components PNS components Shewit Gebre	Organization of the Circulatory System 1. Cardiovascular system Vascular circuits Blood vessels (Arteries, veins, capillaries) Shewit Gebre	Introduction to Histology & Its Methods of Study prof. MEKEBEB A
10:30-10:45 a.m.	BREAK				
10:45:-12: 00 a.m. INSTRUCTOR	Reading assignment: Skeletal system	Muscular system: General description Features and distribution of skeletal muscles Contraction and function of muscles Shewit Gebre	Somatic and autonomic nervous system ANS components Shewit Gebre	Lymphatic vessels and lymph nodes of the body Shewit Gebre	Cytology 1: Cell membrane Prof. MEKEBEB A
12:00-1:30 PM	LUNCH				
1:30-4:30 PM	Reading assignment: Skeletal system	DR 1: Group 1 Mr. yossef DR 2: Group 2	DR 1: Group 3 Dr. SORESSA	DR 1: Group 5 Prof. Grima DR 2: Group 6	Hist. Lab 1:Group 1 1:30- 3:30 PM Hist. Lab 2:Group 2

INSTRUCTOR	system	Dr.Solomon T	DR 2: Group 4 Mr. Abay	Mr.. Fikre Bayu	3:30- 5:30 PM Mr.Yossef
MODULLE ONE: BSOF 1011 THEME ONE TIME TABLE FOR WEEK TWO (Dec 12- Dec 16)					
TIME	MONDAY (Dec 12)	TUESDAY (Dec13)	WEDNESDAY (Dec 14)	THURSDAY (Dec15)	FRIDAY (Dec 16)
9:00:-10: 30 AM INSTRUCTOR	Cytology 1: Cell organeles Prof. MEKEBEB A	Cytology 3: Nucleus & Cell cycle pro. MEKEBEB A	What Physiology means? Cell AbebayeAragaw	Body Fluids AbebayeAragaw	Study period
10:30-10:45 AM	BREAK				
10:45:-12: 00 AM INSTRUCTOR	Cytology 2: Cytoskeletal system and cell deposits prof. MEKEBEB A	Mitosis vs meiosis Dr.Solomon T	Cell AbebayeAragaw	Body Fluids AbebayeAragaw	
12:00-1:30 PM	LUNCH				
1:30-4:30 PM INSTRUCTOR	Hist. Lab 1:Group 3 1:30- 3:30 PM Hist. Lab 2:Group 4 3:30- 5:30 PM Mr.Abay	Hist Lab 1:Group 5 1:30- 3:30 PM Hist Lab 2:Group 6 3:30-5:30 PM Dr.Soresa	Hist Lab 1:Group 1 1:30- 3:30 PM Hist Lab 2:Group 2 3:30-5:30 PM Mr. Fikre	Hist Lab 1:Group 2 1:30- 3:30 PM Hist Lab 2:Group 3 3:30-5:30 PM Mr. yosef	Hist. Lab 1:Group 5 1:30- 3:30 PM Hist. Lab 2:Group 6 3:30- 5:30 PM Mrs. Shewit

MODULLE ONE: BSOF 1011 THEME ONE TIME TABLE FOR WEEK THREE (Dec 19 – Dec 23)					
TIME	MONDAY(Dec19)	TUESDAY(Dec 20)	WEDNESDAY(Dec21)	THURSDAY(Dec22)	FRIDAY (Dec23)

9:00:-10: 30 AM INSTRUCTOR	Test 1 Gross anatomy and Histology (25%)	The common elements of life; Biologically important functional groups; Water and acid base properties.; Buffers; Non-covalent interactions (1h) Carbohydrate Structure: Tinsae	Polysaccharides & glycosaminoglycans (1h) Tinsae	Amino Acids, Peptides and Proteins: Structure and classification of amino acids; SisayAddisu	Amino Acids, Peptides and Proteins: Protein folding and denaturation Nucleotides and Nucleic Acids SisayAddisu
10:30-10:45 AM	BREAK				
10:45:-12: 00 AM INSTRUCTOR		Monosaccharides, sugar derivatives and disaccharides (2h) Tinsae	Lipid Structure: Glycerophospholipids, sphingolipids and cholesterol; Biological Membranes Tinsae	Structural levels and classification of proteins SisayAddisu	Types and structures of RNA and DNA; Denaturation and renaturation of DNA SisayAddisu
12:00-1:30 PM	LUNCH				
1:30-4:30 PM INSTRUCTOR	Free	Hist. Lab 1: Group 1 1:30- 3:30 PM Hist. Lab 2: Group 2 3:30- 5:30 PM Shewit Gebre Biochem lab: Group 5 and 6 Laboratory discipline. Total protein determination;	Hist. Lab 1: Group 3 1:30- 3:30 PM Hist. Lab 2: Group 4 3:30- 5:30 PM Shewit Gebre Biochem lab: Groups 1 and 2 Laboratory discipline. Total protein determination;	Hist. Lab 1: Group 5 1:30- 3:30 PM Hist. Lab 2: Group 6 3:30- 5:30 PM Shewit Gebre Biochem lab: Groups 3 and 4 Laboratory discipline. Total protein determination;	Group work (seminar)

MODULLE ONE: BSOF I THEME ONE TIME TABLE FOR WEEK FOUR (Dec26 – Dec30)					
TIME	MONDAY (Dec26)	TUESDAY(Dec27)	WEDNESDAY(Dec28)	THURSDAY (Dec29)	FRIDAY (Dec30)
9:00:-10: 30 AM INSTRUCTOR	Introduction to Embryology Outlines of Reproductive organs Dr.Solomon T	Transport of gametes Dr.Solomon T	Integrated seminar day	seminar day	Study period
10:30-10:45 AM					

10:45:-12: 00 AM INSTRUCTOR	Gametogenesis Dr.Solomon T				
12:00-1:30 PM					
1:30-4:30 PM INSTRUCTOR	Group work (seminar)	Group work (seminar)			

MODULLE ONE: BSOF I THEME ONE TIME TABLE FOR WEEK FIVE (Jan2-Jan6)					
TIME	MONDAY (Jan2)	TUESDAY (Jan3)	WEDNESDAY (Jan4)	THURSDAY (Jan5)	FRIDAY (Jan 6)
9:00:-10: 30 AM INSTRUCTOR	Test 2 Physiology, Biochem, embryo) (25%)	Study period	Study period	Study period	Final Examination 40%
10:30-10:45 AM					
10:45:-12: 00 AM INSTRUCTOR					
12:00-1:30 PM					
1:30-4:30 PM INSTRUCTOR					

Module

VIII. Reading Resources

1. Sadler, J.W. *Langman's Medical Embryology*, Williams & Wilkins, Baltimore.
2. Moore, K.L. *The Developing Human: Clinically Oriented Embryology* W.B. Saunders, Philadelphia.
3. Junqueira, L.C., Carneiro, J. & Kelley, R.O. *Basic Histology*. Appleton & Lange.
4. Ross MH, Romrell, LJ & Kaye, G.I. *Histology. A Text and Atlas*. Williams & Wilkins. London
5. Fawcett, D.W. *A Text book of Histology*, Chapman & Hall.
6. DiFiore's *Atlas of Histology with Functional Correlations*. Williams & Wilkins. London.
7. Moore KL, Dalley AF. *Clinically Oriented Anatomy*, 6th ed. Baltimore: Lippincott Williams & Wilkins, 2010.
8. Williams, et al. *Grey's Anatomy*, 38th Ed. Church. Livingston, London. 1995
9. Ellis, H. *Clinical anatomy*, 11th ed. Blackwell publishing, 2006.
10. Romanes, G.J., *Cunningham's Text Book of Anatomy* 12th ed.
11. Tortora, GJ & Derrickson B. *Principles of Anatomy and Physiology*, 12th ed. John Wiles & Sons. Inc, 2009.
12. Van De Graff, MK. *Human anatomy*, 5th ed. WCB/McGraw-Hill, 1998.
13. Textbook of Medical physiology, A. Guyton, 10th edition
14. Review of Medical physiology, William F. Ganong, 16th edition
15. Human Physiology, Seven Edition. Vander Sherman Luciano
16. Weaver, Robert F. *Molecular Biology*, 2nd ed. New York : McGraw-Hill, 2002.
17. David L. Nelson and Michael M. Cox *Lehninger Principles of Biochemistry*, 6th edition

IX. Lecture Topics

T h e m e	LECTURE TOPICS Anatomy
1	THEME ONE: Introduction to anatomy (19hrs) 1.1. Definition, history and subdivisions of Anatomy, Approaches to study anatomical position
	1.2 Anatomical , terminologies and body planes, Anatomical variations 1.3. Fascias, bursae and potential spaces (body cavities).
	2. Review of body systems 2.1. Skeletal system 2.1.1. Osteology: Axial and appendicular skeleton 2.1.2. Arthrology:Types of bony Joints
	2.2. Muscular system (myology) 2.2.1. Description of axial skeletal muscles 2.2.2. Description of appendicular skeletal muscles
	3.3. Organization of the nervous system and terminologies 3.3.1. Components of the CNS 3.3.2. Components of the PNS and nerve plexuses 3.3.3. Organization of ANS
	TEAM ONE: Back (16hrs) 3.1. General description; key features; components: skeletal, spinal nerves, surface anatomy; curvatures. 3.2. Typical and atypical vertebrae 3.3. Back ligaments and joints 3.4.Back musculature 3.5. Clinical correlation; Lumbar puncture; disc prolapse (to be taught together with clinicians)
	1.Introduction to Histology & Its Methods of Study 2. Cytology 1: Cell membrane 3. Cytology 2: Protoplasm 4. Cytology 3: Nucleus & Cell cycle
	5.Epithelial Tissue 1: Forms & Characteristics 6. Epithelial tissue 2: Covering & Glandular Epithelium
	7. Connective tissue 1: Connective tissue proper

	8. Connective tissue 2: Special Connective Tissue
	9. Nervous Tissue 1: Neurons 10. Nervous tissue 2: Neuroglia
	11. Peripheral Nervous System
	1. Introduction to Embryology 1. Outlines of the Reproductive Organs and Gametogenesis
	2. First Week of Development Second Week of Development 3. Third Week of Development
	4. Embryonic Period
	5. Fetal Period
	6. Extra-embryonic Membranes and Placenta 7. Multiple Pregnancy
	8. Human Birth Defects I 9. Human Birth Defects I
	10. Development of Body Cavities, Primitive Mesenteries, and the Diaphragm, Malformations 11. Development of the Integumentary System, Malformations
	Physiology
	What Physiology means ---- 1 lecture 1.1. Definition 1.2. Domain and process of physiology 1.3. Functional Categories of Physiology • Synthetic • Analytic 2. The Cell----- 3 lectures 2.3. General objectives(GO) 2.4. Cell Theory 2.5. Cellular Structure 2.6. Chemical Composition of the Cell 2.7. Cell Division and Differentiation 2.8. Cellular Functions 2.9. Communication Between Cells
	3. Body Fluids----- 3 lectures 3.3. General Objectives (G.O) 3.4. Routes of fluid transfer 3.5. Body composition and Body Fluid Compartment 3.6. Measurement of Body fluids 3.7. Distribution of ions between ECF and ICF 3.8. Water balance – Input and output 3.9. Regulation of Body Fluids- Osmotic and Volume Regulation 3.10. Clinical Correlates

	Disturbance of Electrolyte Composition-Defecit and Excess
	4. Mechanism of Physiological Adjustment ----- 6 lectures 4.1. General Objectives (G.O) 4.2. Introduction • The LIFE • Wisdom of the Body • Ontogeny (development) of Control Mechanisms • Types of Phsiological Adjustment 4.3. The Concept of Homeostasis • Definition and General Principles • Historical Background • Modes and Levels of Physiological Regulation • Loop Systems • Feedback Mechanism • Feedforward Mechanism 4.4. The Concept of Allostasis • Definition and Principles • Allostatic Load • Types of Allostatic Overload Indicators of Allostatic Overload
	4.5. The Concept of Adaptation (Heterostasis) • Definition • Adaptation to Climatic Changes Adjustment to High Altitude
	5. Autonomic Nervous System 5.1. Introduction to ANS Physiologic anatomy of ANS Basic characteristics of the SANS and PANS Cholinergic and adrenergic fibers Mechanisms of transmitter secretion and removal Receptors of the effector organs Acetylcholine (muscarinic, nicotinic) Adrenergic receptors (alpha and beta) 5.2. The presynaptic control of transmitter release Effects on specific organs Function of the adrenal medulla Activity pattern of ANS Sympathetic and parasympathetic tone Effects of autonomic denervation Denervation supersensitivity Autonomic functions after spinal cord transaction
	6. Autonomic functions after spinal cord transaction

	TEAM ONE: Biochemistry
	The common elements of life; Biologically important functional groups; Water and acid base properties.; Buffers; Non-covalent interactions (1h) <i>Carbohydrate Structure:</i> Monosaccharides, sugar derivatives and disaccharides (2h) Demonstration: Ball and stick models of organic molecules; measurement of pH in biological fluids
	<i>Carbohydrate Structure :</i> Polysaccharides and glycosaminoglycans (1h) <i>Lipid Structure:</i> Fatty acids and triacylglycerols (1h) <i>Lipid Structure:</i> Glycerophospholipids, sphingolipids and cholesterol; Biological Membranes
	<i>Amino Acids, Peptides and Proteins:</i> Structure and classification of amino acids; Acid-base properties of amino acids; Structural levels and classification of proteins
	<i>Amino Acids, Peptides and Proteins:</i> Protein folding and denaturation (1h) <i>Nucleotides and Nucleic Acids</i> Purines and pyrimidines; the types and structures of RNA and DNA; denaturation and renaturation of DNA (2h)